KQC7016 DATA ANALYTICS

SEMESTER 2, 2025/2026

LAB 2 REPORT

Github Link: https://github.com/u2103136-lab/KQC7016-Data-Analytics/tree/main/Lab%202

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1. Introduction

The WorldEnergy.csv dataset contains country-level energy data from 1965 to 2024, including primary energy consumption, renewable and fossil fuel shares, electricity mix, and consumption by source (solar, wind, hydro, nuclear). In Lab 2, we focus on six Southeast Asian countries (Indonesia, Malaysia, Philippines, Singapore, Thailand, Vietnam) alongside global trends (World). Our objective is to understand regional energy transitions, compare renewable versus fossil fuel dependence, and identify relationships among energy variables through exploratory data analysis.

2. Data Analytics Projects

- **Descriptive & Comparative Analysis:** Track energy consumption trends over time and compare renewable shares across countries or income groups.
- **Hypothesis Testing (e.g., ANOVA, t-tests)**: Determine whether factors such as GDP level, region, or time period significantly affect renewable adoption or low-carbon energy use.
- **Correlation & Predictive Modelling**: Explore relationships between energy variables (e.g., GDP vs. renewable share) and forecast future energy consumption or renewable growth.

3. Code Explanation

3.1 Import Libraries and Set Up Colour Palette

```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns
import warnings
warnings.filterwarnings('ignore')

COUNTRY_COLORS = {
    'World': '#2C3E50', 'Indonesia': '#E74C3C', 'Malaysia': '#27AE60',
    'Philippines': '#3498DB', 'Singapore': '#E67E22', 'Thailand': '#F1C40F', 'Vietnam':
'#9B59B6'
}
COUNTRIES = list(COUNTRY_COLORS.keys())
SELECTED = [c for c in COUNTRIES if c != 'World']
sns.set_theme(style='whitegrid', font_scale=1.05)
```

Essential libraries for data handling, plotting, and suppressing warnings. A custom colour dictionary ensures consistent colours for each country. SELECTED excludes ‘World’ for country-specific plots. Seaborn’s whitegrid style improves readability.

3.2 Load and Prepare Data

```
df_raw = pd.read_csv('WorldEnergy.csv')
CORE_VARS = ['country', 'year', 'population', 'gdp', 'primary_energy_consumption',
             'renewables_consumption', 'fossil_fuel_consumption',
'renewables_share_energy',
             'fossil_share_energy', 'solar_consumption', 'wind_consumption',
             'hydro_consumption', 'nuclear_consumption']
df = df_raw[df_raw['country'].isin(COUNTRIES)][CORE_VARS].copy()
df = df[df['year'] >= 1965].reset_index(drop=True)
```

Reads the CSV, selects only the rows belonging to the specified countries, and keeps only the necessary columns. Years before 1965 are discarded because data reliability improves from that point onward.

3.3 Tabular Summary

```
desc_cols = ['primary_energy_consumption', 'renewables_share_energy',
             'fossil_share_energy', 'renewables_consumption', 'fossil_fuel_consumption']
print(df[df['country'].isin(SELECTED)][desc_cols].describe().round(2))
```

Prints descriptive statistics (mean, std, min, max, quartiles) for five key energy variables across the selected Southeast Asian countries, giving an overview of the data distribution.

3.4 World Energy Trends

```
world_data = df[df['country'] == 'World'].dropna(subset=['primary_energy_consumption'])
fig, (ax1, ax2) = plt.subplots(1, 2, figsize=(16, 6))
ax1.plot(world_data['year'], world_data['primary_energy_consumption'], color='#2C3E50',
lw=2.5)
ax1.set_title('Global Primary Energy Consumption')
ax2.fill_between(world_data['year'], world_data['fossil_share_energy'], color='#E74C3C',
alpha=0.5, label='Fossil Fuel Share')
ax2.fill_between(world_data['year'], world_data['renewables_share_energy'],
color='#27AE60', alpha=0.7, label='Renewable Share')
ax2.legend()
plt.show()
```

Two subplots, a Global primary energy consumption over time, a line plot showing the long-term upward trend and an area chart comparing fossil fuel share vs renewable share of global primary energy, highlighting the slow but steady rise of renewables.

3.5 Southeast Asia: Renewable & Fossil Shares over Time

```
fig, (ax1, ax2) = plt.subplots(1, 2, figsize=(16, 6))
for country in SELECTED:
    c_df = df[df['country'] == country]
    ax1.plot(c_df['year'], c_df['renewables_share_energy'], label=country,
color=COUNTRY_COLORS[country], lw=2)
    ax2.plot(c_df['year'], c_df['fossil_share_energy'], label=country,
color=COUNTRY_COLORS[country], lw=2)
ax1.set_title('Renewable Share (%) per Country')
ax2.set_title('Fossil Share (%) per Country')
```

```
ax2.legend(bbox_to_anchor=(1.05, 1))
plt.show()
```

Multi-line plots for each of the six SEA countries. The left plot shows renewable share over time; the right shows fossil fuel share. These visualisations reveal how each country's energy mix has evolved.

3.6 Bar Chart & Boxplot

```
latest_yr = df['year'].max()
latest_data = df[(df['year'] == latest_yr) &
(df['country'].isin(SELECTED))].sort_values('renewables_consumption')
fig, (ax1, ax2) = plt.subplots(1, 2, figsize=(16, 6))
ax1.barh(latest_data['country'], latest_data['renewables_consumption'],
color=[COUNTRY_COLORS[c] for c in latest_data['country']])
ax1.set_title(f'Renewable Energy Consumption (TWh) – {latest_yr}')
sns.boxplot(data=df[df['country'].isin(SELECTED)], x='country',
y='renewables_share_energy', palette=COUNTRY_COLORS, ax=ax2)
ax2.set_title('Historical Renewable Share Distribution')
plt.xticks(rotation=45)
plt.show()
```

Horizontal bar chart of absolute renewable consumption in the most recent year, ranking countries. Boxplot showing the distribution (median, quartiles, outliers) of renewable share over the entire 1965-2024 period for each country.

3.7 Correlation Heatmap (World only)

```
heatmap_cols = ['primary_energy_consumption', 'renewables_consumption',
'fossil_fuel_consumption',
                'renewables_share_energy', 'fossil_share_energy', 'solar_consumption',
                'wind_consumption', 'hydro_consumption']
world_hm = df[df['country'] == 'World'][heatmap_cols].dropna()
corr = world_hm.corr()
sns.heatmap(corr, annot=True, fmt='.2f', cmap='RdYlGn', mask=np.triu(np.ones_like(corr,
dtype=bool), k=1))
plt.show()
```

For global data, a correlation matrix is computed and displayed as a heatmap. Only the lower triangle is shown. Positive correlations in green, negative in red. This helps identify strong relationships (e.g., fossil fuel consumption vs. primary energy).

3.8 World Renewable Mix Stacked Area

```
mix_cols = ['solar_consumption', 'wind_consumption', 'hydro_consumption']
world_mix = df[df['country'] == 'World'][['year'] + mix_cols].dropna()
plt.stackplot(world_mix['year'], world_mix['hydro_consumption'],
world_mix['wind_consumption'], world_mix['solar_consumption'],
              labels=['Hydro', 'Wind', 'Solar'], colors=['#3498DB', '#27AE60', '#F39C12'],
alpha=0.85)
plt.legend()
plt.show()
```

Stacked area chart showing the absolute consumption of hydro, wind, and solar, the three main renewable sources. Hydro dominates, but wind and solar have grown rapidly since the 2000s.

3.9 Univariate Analysis: World Renewables Share

```
world_u = df[df['country'] == 'World'].dropna(subset=['renewables_share_energy'])
fig, axes = plt.subplots(1, 2, figsize=(12, 4))
axes[0].hist(world_u['renewables_share_energy'], bins=20, color='#27AE60',
edgecolor='white')
axes[0].axvline(world_u.mean(), color='red', linestyle='--', label='Mean')
axes[0].axvline(world_u.median(), color='navy', linestyle=':', label='Median')
axes[0].legend()
sns.kdeplot(world_u, color='#27AE60', fill=True, ax=axes[1])
axes[1].text(0.05, 0.9, f'Skewness: {world_u.skew():.2f}\nKurtosis:
{world_u.kurtosis():.2f}', transform=axes[1].transAxes)
plt.show()
```

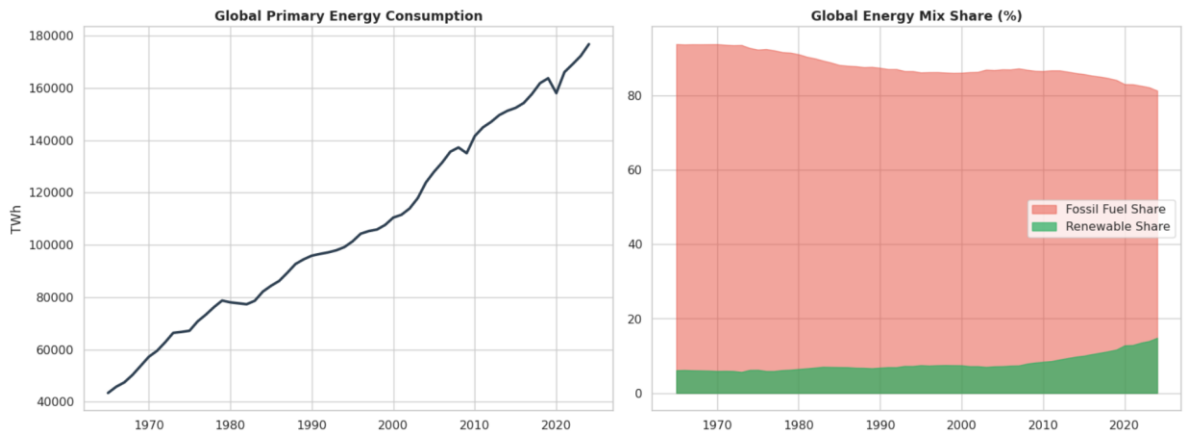
Histogram and kernel density estimate (KDE) for the World's renewable share across all years. Mean and median are marked; skewness and kurtosis describe the distribution shape (right-skewed, peaked).

3.10 Per-Country Summary Table

```
summary_cols = ['country', 'year', 'primary_energy_consumption', 'renewables_share_energy',
                'fossil_share_energy', 'renewables_consumption', 'fossil_fuel_consumption']
latest_all = []
for c in COUNTRIES:
    cdf = df[df['country'] == c].dropna(subset=['renewables_share_energy'])
    if not cdf.empty:
        latest_all.append(cdf.iloc[-1][summary_cols])
summary_df = pd.DataFrame(latest_all).set_index('country')
print(summary_df.round(2))
```

Loops through each country, finds the most recent year with non-missing renewable share data, and prints a table with the latest values for the core variables, a concise snapshot of current energy profiles.

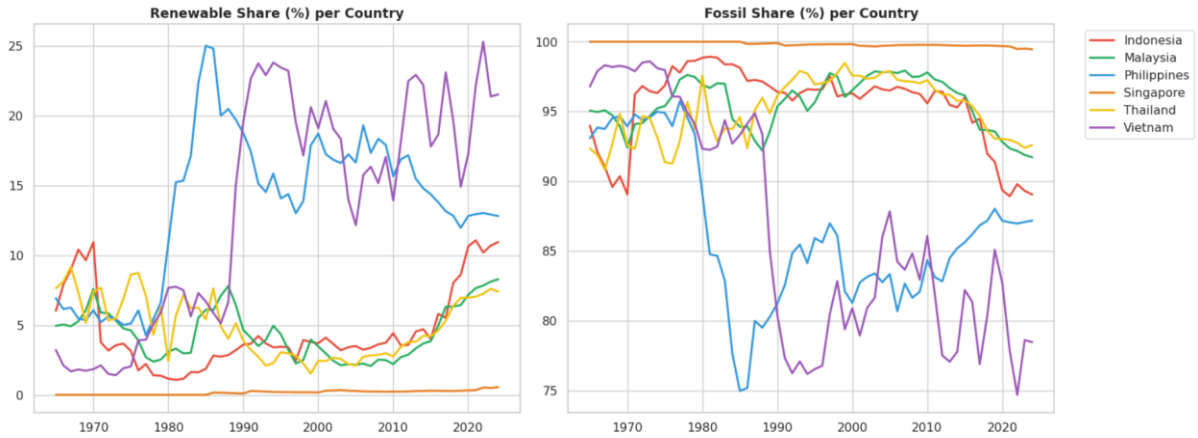
4. Results and Discussion



Key Observation

1. Total global primary energy consumption has roughly quadrupled since 1965, showing a steep and relatively continuous upward trajectory approaching 180,000 TWh.
2. While there are visible dips in total consumption corresponding to major global events (e.g., the ~2009 financial crisis and the 2020 pandemic), they are short-lived, with demand quickly recovering and exceeding previous peaks.
3. Despite the rise of alternative energy, fossil fuels have persistently dominated the global energy mix, consistently accounting for over 80% of total share throughout the timeline.
4. The share of renewable energy is relatively small but shows a clear, accelerating upward trend over the last 10–15 years, climbing toward the 15% mark.

Because total global energy consumption (left graph) is growing so rapidly, a slight percentage decrease in the fossil fuel share (right graph) still means the **absolute volume** of fossil fuels burned globally continues to increase alongside renewables.

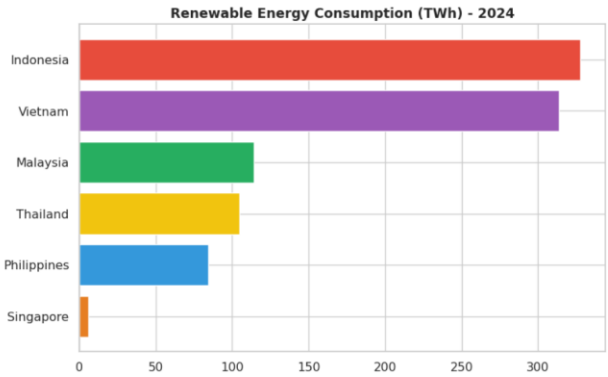


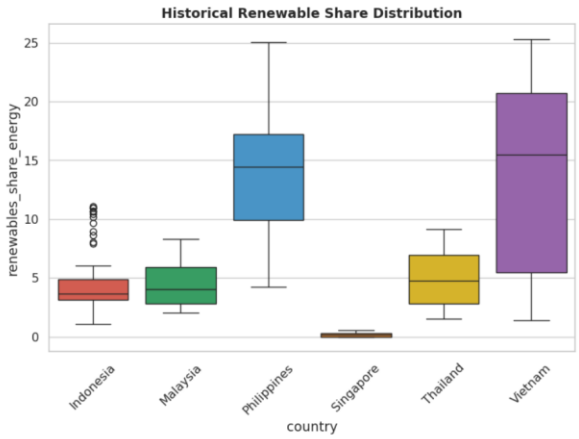
Key Observation

1. Singapore (orange line) exhibits a near 100% reliance on fossil fuels for its entire history, with its renewable energy share remaining practically flat at 0%.
2. Vietnam (purple line) shows the most dramatic and volatile shifts. Its renewable share surged massively in the late 1980s/early 1990s (likely due to large hydroelectric projects), correspondingly dropping its fossil share to the lowest in the region, though it fluctuates wildly.
3. The Philippines (blue line) was the regional leader in renewables during the 1980s, peaking around 25%. However, its renewable share has been in a steady, decades-long decline ever since, with its fossil fuel dependency climbing back up.
4. Indonesia, Malaysia, and Thailand have historically maintained low renewable shares (generally under 10%). However, all three show a synchronized, gradual upward trend in renewables (and a dip in fossil share) starting around 2015.
5. The peaks and valleys in the "Renewable Share" chart perfectly mirror the "Fossil Share" chart. This zero-sum relationship indicates that for these nations, any gain in

renewables is directly displacing fossil fuels, with other energy sources (like nuclear) being virtually non-existent.

Conclusion: While individual nations like Vietnam and the Philippines have experienced dramatic, distinct historical shifts in their energy mixes, Southeast Asia as a whole remains heavily entrenched in fossil fuel dependency. However, a slow but unified regional transition toward renewable energy appears to have begun in the last decade.

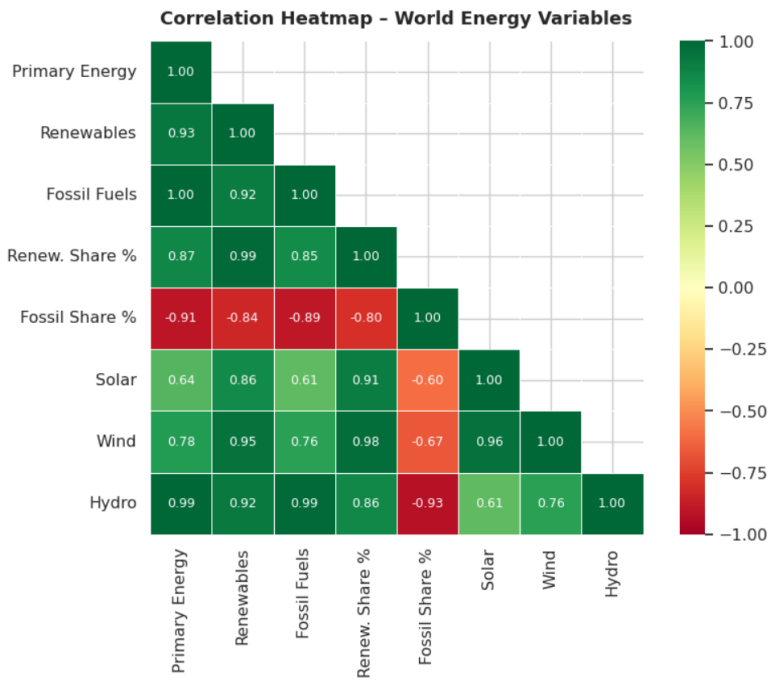




Key Observations:

1. **Indonesia's Volume vs. Share Disparity:** Indonesia leads in absolute renewable energy consumption for 2024 (>300 TWh) but has a very low historical median renewable share (under 5%), indicating its massive total energy demand dwarfs its green energy output.
2. **Vietnam's High Volatility:** Vietnam has the widest historical distribution (ranging from near 0% to 25% share), showing immense historical volatility, despite being the second-largest consumer of absolute renewable energy in 2024.
3. **Philippines' High Share, Low Volume:** The Philippines has one of the highest historical median renewable shares (around 14.5%) but ranks near the bottom in absolute consumption in 2024, suggesting a much smaller overall national energy grid.
4. **The Middle-Tier Twins:** Malaysia and Thailand display remarkably similar profiles. Both have moderate absolute consumption (~100-110 TWh) and tight, consistently low historical renewable shares (medians around 4-5%).

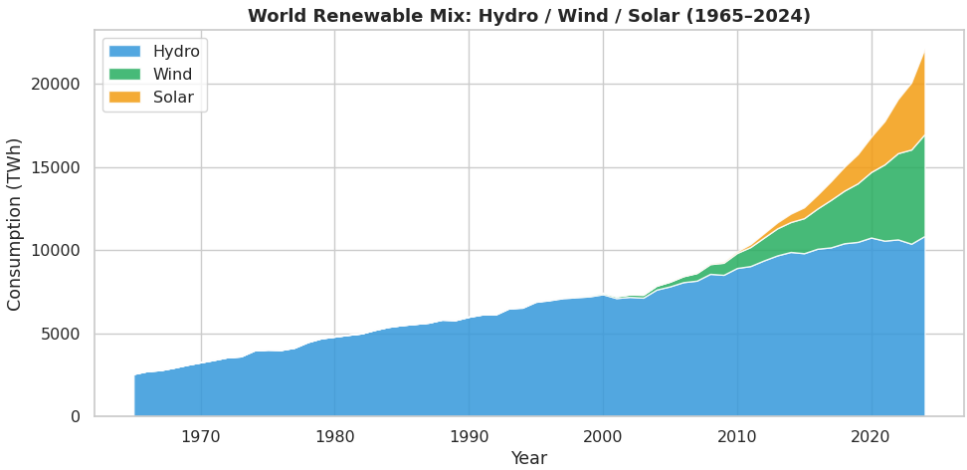
Conclusion: There is a stark disconnect between absolute renewable energy generation and actual grid decarbonization. Large nations like Indonesia produce massive amounts of renewable energy but remain heavily diluted by fossil fuels, whereas countries with smaller overall energy grids like the Philippines achieve a much greener energy mix with significantly less total renewable volume.



Key Observations:

1. **Perfect Energy Coupling:** There is a perfect positive correlation (1.00) between "Primary Energy" and "Fossil Fuels," meaning historical global energy demand has grown in exact lockstep with fossil fuel consumption.
2. **Simultaneous Absolute Growth:** "Renewables" strongly correlate with both "Primary Energy" (0.93) and "Fossil Fuels" (0.92). This indicates that as total energy consumption increases, the absolute volumes of both fossil fuels and renewables are growing together.
3. **Wind and Solar Drive the Mix:** "Wind" (0.98) and "Solar" (0.91) have extremely high correlations with the "Renew. Share %". This shows that these two specific technologies are the primary engines driving the shift in the global renewable energy percentage.
4. **Hydroelectric as Baseline Growth:** "Hydro" correlates almost perfectly with "Primary Energy" (0.99) but much weaker with "Solar" (0.61). This suggests hydroelectric power grew steadily alongside historical baseline energy demand, rather than participating in the recent, rapid spike of new green technologies.

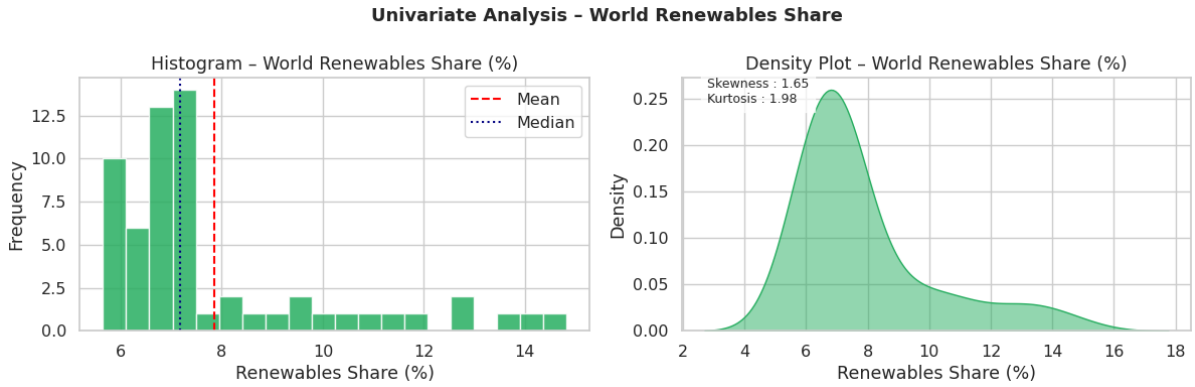
Conclusion: While the total absolute volume of global energy remains fundamentally tied to fossil fuel consumption, the *composition* of the energy mix is actively changing. The transition toward a higher renewable share is being overwhelmingly driven by recent, synchronized surges in wind and solar adoption rather than legacy sources like hydro.



Key Observations:

1. **Historical Hydro Dominance:** For the first four decades shown (1965 to roughly 2005), hydroelectric power was almost exclusively responsible for the world's renewable energy mix.
2. **Linear Hydro Growth:** Hydroelectric consumption has grown at a very steady, linear pace over the entire 60-year period, establishing a massive baseline of around 10,000 TWh by 2024.
3. **The Rise of Wind:** Wind energy begins to make a visible impact on the global mix in the early 2000s, expanding rapidly to become the second-largest source in this group.
4. **Recent Solar Explosion:** Solar energy is the latest entrant, only becoming distinctly visible after 2010. However, it displays the sharpest expansion rate in the most recent years.
5. **Post-2010 Acceleration:** The total combined consumption of these three sources shifts from a gentle slope to a steep, upward curve beginning around 2010, entirely driven by the additions of wind and solar.

Conclusion: The global renewable energy landscape has fundamentally shifted in the 21st century. While hydroelectricity provides a massive and steady foundation, the recent, explosive growth in total renewable energy consumption is almost exclusively powered by the rapid adoption of wind and solar technologies.



Key Observations:

1. **Right-Skewed Distribution:** The data exhibits a strong positive (right) skew, quantitatively confirmed by a skewness value of 1.65.
2. **Mean Exceeds Median:** Due to the rightward skew, the mean (red dashed line) is pulled significantly higher than the median (blue dotted line).
3. **Historical Concentration:** The overwhelming majority of the data points are tightly clustered at the lower end, peaking sharply in the 6% to 8% range.
4. **The Long Tail:** There is a sparse but extended right tail stretching up to 15%, representing a few anomalous data points (recent years) that break from the historical norm.
5. **Emerging Secondary Shift:** The density plot shows a faint secondary "bump" or plateau in the 12%–15% range, indicating a distinct cluster of recent high-adoption years.

Conclusion: For the vast majority of the observed timeline, the global renewable energy share has been heavily stagnant, trapped in a narrow 6–8% band. The strong rightward skew and extended tail visually capture a recent, aggressive breakaway from this historical baseline as renewable adoption finally accelerates.

5.0 Conclusion

While global and Southeast Asian energy demands continue to surge alongside absolute fossil fuel consumption, a definitive structural transition is underway. Historically stagnant, the global renewable share is now accelerating rapidly, driven by explosive recent growth in wind and solar. Regionally, Southeast Asia remains deeply entrenched in fossil fuels, exhibiting a stark disconnect between high absolute renewable generation and actual grid decarbonization. However, a unified regional shift toward green energy has clearly emerged over the last decade. Ultimately, while total fossil fuel volumes remain stubbornly high, the proportional shift toward a renewable-driven energy future is undeniable.